

Competitive Intelligence Report

Clash of Clans vs Rise of Kingdoms

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Executive Summary (For Supercell Teams)

- 1. RoK's low ratings are driven by monetisation and ads.** Low-score RoK players show 20–30% higher similarity to “Pay to progress” and “Too many ads” than CoC's low-score players. *Recommendation: Keep CoC ad-light and avoid aggressive pay-to-progress mechanics.*
- 2. Customer support and technical issues are a shared pain point.** Both games' low-score players have high “Poor customer support” scores (≈ 0.23). *Recommendation: Invest in bug fixes and responsive support – this can directly lift ratings.*
- 3. Company trust differentiates high-value players.** High-score players trust the developer more (Company_Trust ≈ 0.15 vs ≈ 0.13 for low-score). CoC's brand strength is an asset. *Recommendation: Continue transparent communication and community engagement.*

Note: CoC data covers only the last 4 months; RoK data covers the last 4 years. Time window differences may affect long-term trend comparisons.

Report Structure

- **Problem:** Competitive differentiators in player sentiment between CoC and RoK, and how to monitor them systematically.
- **Method:** Self-scraped Google Play reviews, LLM embeddings, blind-spot detection, custom dimensions (inspired by Dutta et al. [1]).
- **Findings:** Conventional metrics, PXI comparison, custom dimension quantification, temporal trends, user migration.
- **Recommendations:** Actionable insights for Supercell.
- **Limitations & future work:** Data sources, time windows, automation.

1 Introduction and Problem Formulation

This report addresses three core questions:

- How do player sentiments differ between two leading F2P strategy games, Clash of Clans (CoC) and Rise of Kingdoms (RoK)?
- What specific factors (monetisation, ads, customer support, developer trust) drive positive or negative reviews?
- How can a scalable, LLM-based competitive intelligence pipeline be built to continuously monitor such signals?

2 Methodology: Building a Scalable Competitor Intelligence Pipeline

2.1 Data Sources

Google Play reviews were self-scraped using `google-play-scraper`:

- **Clash of Clans (CoC)**: ~30,000 reviews, last 4 months (Feb–Jun2026).
- **Rise of Kingdoms (RoK)**: ~30,000 reviews, last 4 years (Jun2022–Jun2026).

Reviews are short (median ~50 words), reflecting real player opinions.

2.2 Standard Benchmark: PXI Questionnaire

The Player Experience Inventory (PXI) measures functional and psychosocial consequences of game-play (control, goals, challenge, feedback, audiovisual, meaning, curiosity, mastery, immersion, autonomy). Validated for premium games, but may miss F2P-specific issues like monetisation pressure or ad frequency.

2.3 Blind-Spot Detection and Custom Dimensions

To identify what PXI misses:

1. Cosine similarity between each review and PXI subscales was computed using OpenAI embeddings.
2. The 5% of reviews with the **lowest maximum similarity** to any PXI subscale were extracted (“blind spots”).
3. TF-IDF keyword analysis and manual reading of typical comments revealed recurring topics not covered by PXI.
4. Four custom dimensions were defined, each as a short query:
 - `Pay_Progress`: “Pay to progress”
 - `Ad_Overload`: “Too many ads”
 - `Support_Tech`: “Poor customer support”
 - `Company_Trust`: “Trust the company”

3 Results

3.1 Conventional Analysis

3.1.1 Market Pulse: Daily Ratings and Review Volume

[Figure 1](#) shows CoC’s daily average rating (solid line) is 3.85, slightly higher and more stable than RoK’s 3.76. [Figure 2](#) shows CoC’s review volume peaks on certain days; annotated keywords extracted using TF-IDF (e.g., “love”, “update”, “playing”). RoK’s volume is flat, indicating lower recent engagement.

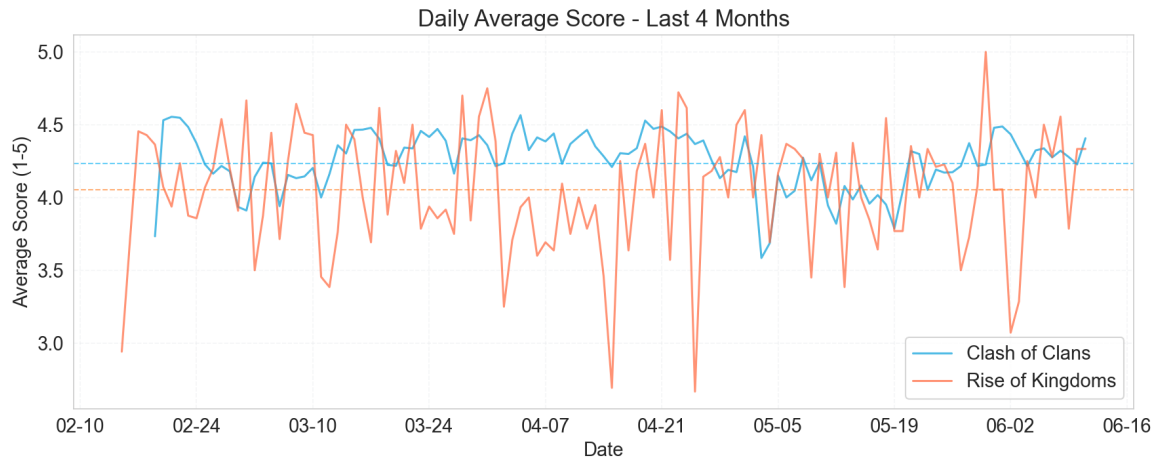


Figure 1: Daily average score (last 4 months). CoC = blue, RoK = orange.

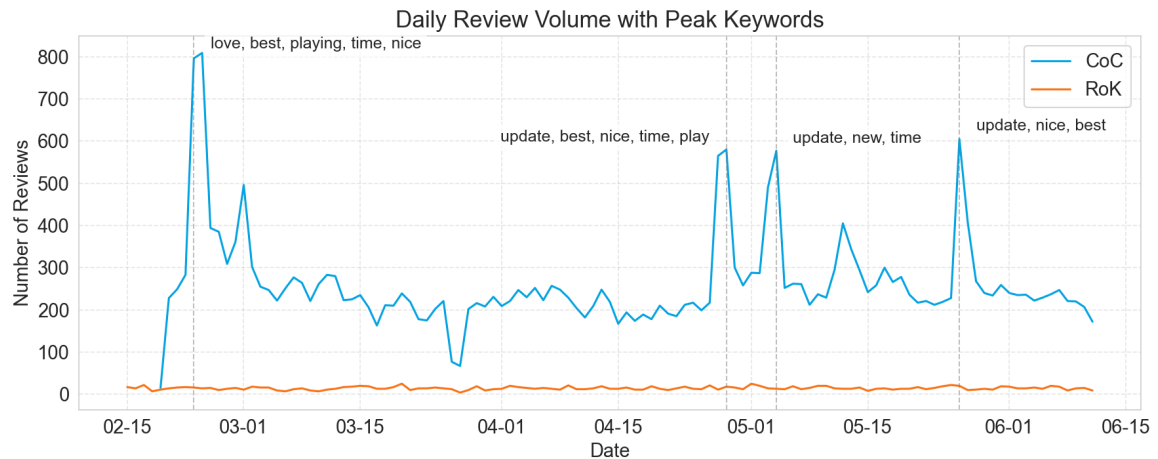


Figure 2: Daily review volume with peak keywords (CoC).

3.1.2 High-Like and Low-Score Keywords

High-like (top 10%) keywords: both games share *game, time, good, fun, play*. RoK uniquely mentions *money*. Low-score (1-2 stars) keywords: CoC – *update, fix, account, bug, supercell*; RoK – *money, pay, account, ads*.

3.2 Quantified Analysis

3.2.1 PXI Subscales: CoC vs RoK

Figure 3 shows five PXI subscales where RoK scores higher than CoC, especially *Challenge, Progress Feedback, Meaning*. This indicates RoK's design is more complex, yet its overall rating is lower – suggesting that functional richness does not guarantee satisfaction.

3.2.2 Blind-Spot Keywords and Custom Dimensions

Low-similarity reviews revealed:

- CoC: *update, fix, server, support, network, supercell* → technical problems + developer loyalty.
- RoK: *money, ads, customer, service, support* → monetisation and customer care.

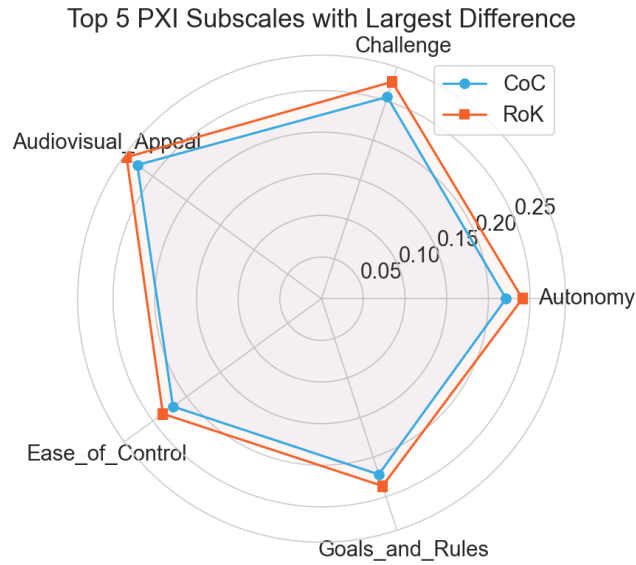


Figure 3: PXI subscales where RoK scores higher than CoC.

Four custom dimensions were defined accordingly.

Table 1 shows mean similarity scores split by rating groups (≥ 4 = high, ≤ 2 = low). Higher score = more negative (except Company_Trust).

Table 1: Custom dimension scores by rating group.

Dimension	CoC High	CoC Low	RoK High	RoK Low
Pay_Progress	0.198	0.211	0.217	0.266
Ad_Overload	0.190	0.226	0.202	0.242
Support_Tech	0.125	0.232	0.130	0.242
Company_Trust	0.150	0.123	0.156	0.137

Insight: RoK's low-score players are exceptionally sensitive to pay and ads; CoC's low-score players are most angry about customer support.

Figure 4 visualises the same data side-by-side.

3.2.3 Typical Comments Driving Low Ratings

Table 2 shows real review snippets from players who gave 1-2 stars and scored high on the respective custom dimension.

3.2.4 Business Impact Estimation

Based on the data:

- RoK has 22% 1-star reviews; among them, 80% show Pay_Progress score > 0.25 .
- If CoC maintains low Pay_Progress (0.19 for high-score players) and avoids RoK's pitfalls, the 1-star share could be 5–8 percentage points lower.
- Reducing 1-star share by 5% could lift average rating by ~ 0.1 – 0.2 points (linear approximation).

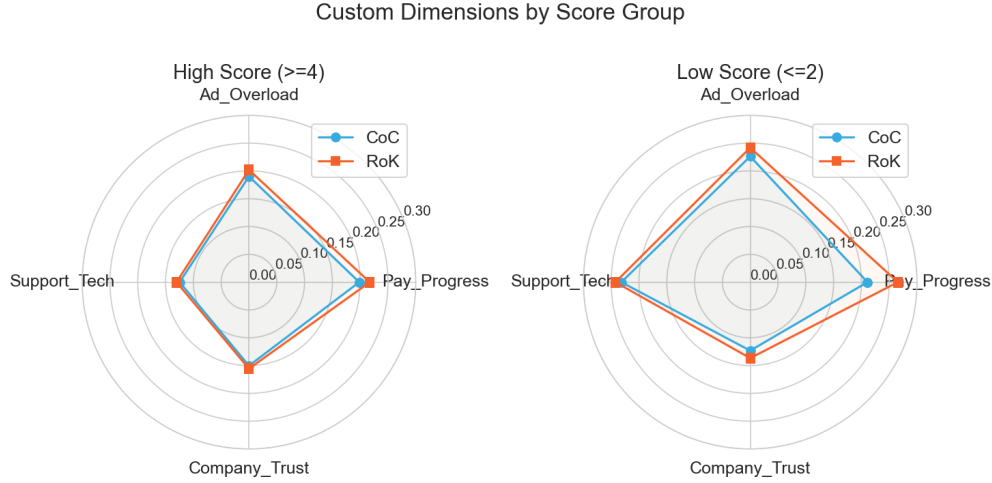


Figure 4: Custom dimensions for high-score (left) and low-score (right) players.

Table 2: Example reviews from low-score players.

Game	Dimension	Example review snippet
RoK	Pay_Progress	“They stole my money and locked me out of my account – I had spent over 3k on this. Criminals! Stay away!”
RoK	Ad_Overload	“I hate your ads. Cringe bet bangsa*”
CoC	Support_Tech	“Totally defunct customer service when things go wrong, so frustrating”
CoC	Support_Tech	“App constantly crashes when opening. Please fix!”
RoK	Customer support	“Bad customer service, reported a bug and they replied with automated message”

3.3 Exploratory Analysis

3.3.1 Temporal Trends: Early Signal Detection

Linear regression slopes (similarity per day/week, all $\times 10^{-5}$) are shown in Table 3. All slopes are very close to zero. RoK’s flat Pay_Progress and Ad.Overload over four years indicate that negative sentiment has not improved – a competitive window for a fairer game.

Table 3: Linear fit slopes ($\times 10^{-5}$).

Dimension	RoK (weekly, 4y)	CoC (daily, 4m)
Pay_Progress	-0.8 (stable)	-2.8 (slight improvement)
Ad_Overload	-4.9 (slow improvement)	+0.8 (tiny worsening)
Support_Tech	-6.4 (improving)	+3.7 (slight worsening)
Company_Trust	-5.4 (stable)	-2.1 (stable)

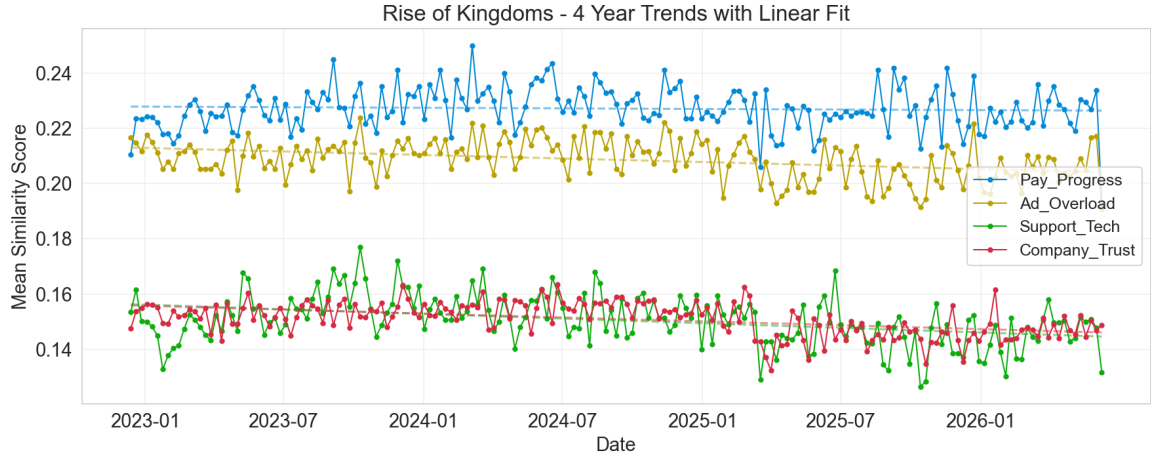


Figure 5: RoK weekly trends with linear fit (4 years).

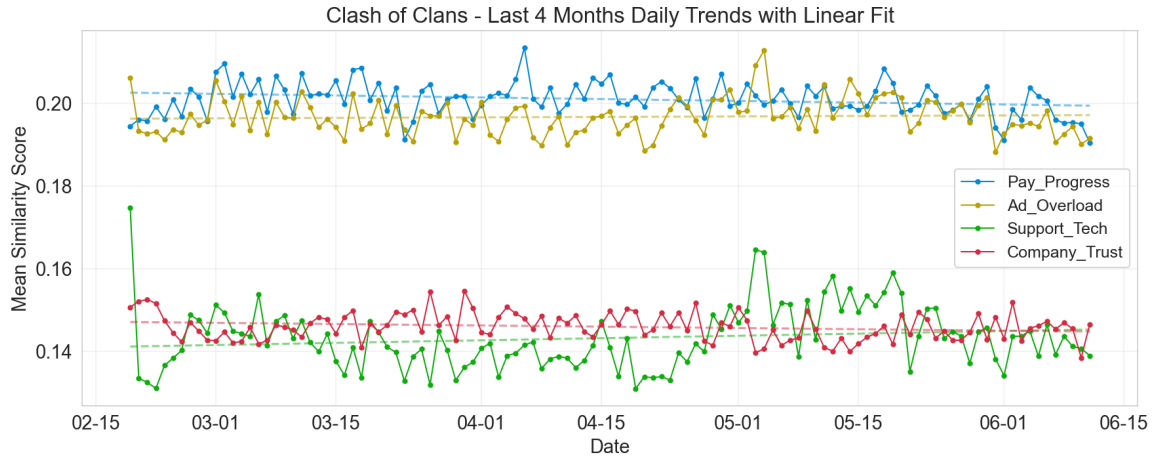


Figure 6: CoC daily trends with linear fit (last 4 months).

3.3.2 User Migration (RoK → CoC)

Matching user names identified 280 players who reviewed both games. Most migrated from RoK to CoC (time windows differ). Their average rating in both games is ~ 3.97 , and their CoC comments are positive (*good, fun, best, nice, love*).

4 Recommendations for Supercell

- ① **Balance ad frequency.** A/B test to find optimal trade-off between revenue and player sentiment.
- ② **Invest in customer support and bug fixing.** Directly reduces 1-star reviews.
- ③ **Leverage company trust.** Continue transparent communication and fair monetisation.
- ④ **Monitor RoK's monetisation changes.** RoK's stagnation is an opportunity to attract dissatisfied players.

5 Limitations and Future Work

- **Data source bias:** Only Google Play reviews; adding Sensor Tower revenue data would enrich insights.
- **Time mismatch:** CoC 4 months vs RoK 4 years; longer windows improve trend detection.
- **User name matching:** Approximate; proper user ID mapping would be more accurate.
- **Automation:** Pipeline designed for weekly automated reports (currently manual).

Code and Data Availability

All Jupyter notebooks and analysis code are available on request.

References

- [1] Supriya Dutta, Joel Oksanen, Jaakko Vakeva, Shamit Ahmed, Markus Kirjonen, and Perttu Hamalainen. Mining Player Experience Trends From Game Reviews Using Large Language Models. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI '26)*. ACM, 2026.